APURBA DEBNATH

apurba.debnath@ucsf.edu | linkedin.com/in/apurba-debnath | apurba.github | apurba.github.io

EDUCATION

University of California, San Francisco — San Francisco, CA
Master of Science in Biomedical Imaging | GPA: 3.9/4.0 — SEP 2025 – SEP 2026
Thesis: *Spectral Graph modeling of abnormal neural synchronizations in Alzheimer's disease.*

National Institute of Technology Durgapur — Durgapur, India
Bachelor of Technology in Biotechnology | Major GPA: 3.78/4.00, CS GPA: 4.0/4.0. — DEC 2020 – MAY 2024
Thesis: *Machine learning-based approach for the design and release kinetics prediction of the PLGA-based drug delivery systems.*

RESEARCH EXPERIENCE

University of California, San Francisco, San Francisco, CA, USA — JAN 2026 – Present
Graduate Student Researcher, Brain Networks Laboratory, UCSF
Advisors: Prof. Ashish Raj & Prof. Srikantan Nagarajan

- Conducted signal processing and spectral analysis of EEG/MEG recordings from large-scale cohorts of healthy individuals and patients with Alzheimer's disease (AD).
- Implemented two-step optimization algorithm using the spectral graph model to deconvolve scale-free neural driving signals and model oscillatory brain activity in healthy and AD cohorts, achieving >0.90 Pearson correlation and <0.5 RMSE between modeled and empirical MEG spectra across 246 cortical/subcortical regions while recovering latent scale-free neural driving signals.
- Quantified deconvolved scale-free neural driving signals and demonstrated that AD pathology (FDG-uptake, $A\beta$ and tau accumulation) is associated with significant alterations in scale-free neural dynamics, with spatial alterations across the delta–theta, alpha, and beta bands recapitulating regions selectively vulnerable to AD pathology.
- Demonstrated that altered scale-free neural driving signals are associated with disrupted excitation–inhibition (E/I) dynamics, with AD exhibiting significantly increased excitatory time constants; additionally, greater cognitive decline was associated with significantly increased time constants of long-range excitatory neurons.

– Research area: *Alzheimer's disease, Computational Neuroscience.*

Indian Institute of Technology Madras, Chennai, India — JUL 2024 – JUL 2025
Post-Baccalaureate Researcher, Centre for Integrative Biology and Systems Medicine (IBSE)
Co-Advisors: Dr. Parul Verma (IIT Madras) & Dr. Marie-Constance Corsi (Paris Brain Institute)

- Processed and analyzed EEG/MEG recordings from a longitudinal cohort of 19 healthy participants across four motor imagery (MI)-based BCI training sessions, along with an independent EEG dataset of 16 participants trained using a separate BCI paradigm.
- Developed a biophysical neural mass model with explicit modeling of scale-free brain activity to investigate the neural mechanisms underlying MI-BCI learning, achieving a Pearson correlation of >0.90 and an RMSE of <0.3 between modeled and empirical M/EEG power spectra across most brain regions, subjects, and sessions under both MI and resting-state (RS) conditions.
- Demonstrated regional activation accompanied by a significant reduction in excitatory time constant during MI compared with RS, and revealed that BCI training drives a shift from diffuse to targeted sensorimotor recruitment. Additionally, identified a strong association between the excitatory time constant and MI-related desynchronization/synchronization, supporting its potential as a biomarker of MI-BCI learning and informing the development of biologically informed, individualized BCI training protocols.

– Research area: *Brain-Computer Interface, Computational Neuroscience.*

National Institute of Technology Durgapur, India — AUG 2023 – MAY 2024
Undergraduate Researcher, Department of Biotechnology
Advisor: Prof. Dalia Dasgupta Mandal

- Reviewed and analyzed mathematical models from the literature to characterize drug release kinetics, and implemented 8 ML regression models to predict drug release kinetics in PLGA-based drug delivery systems using a dataset of 1,914 experimental drug release profiles across 20 drug–polymer formulations involving anticancer and non-anticancer drugs.
- Designed an end-to-end predictive modeling pipeline, with Random Forest and XGBoost demonstrating strong predictive performance for early-stage drug release kinetics (up to $\approx 50\%$ cumulative drug release).
- Optimized Random Forest model achieved a test R^2 of 0.71, MSE of 0.0264, and MAE of 0.1169, while reducing the feature space from 17 to 7 key physicochemical features and identifying molecular weight, drug loading capacity, surface-area-to-volume ratio, and time as key determinants of drug release kinetics.

– Research area: *Drug Delivery Systems, Machine Learning*

PUBLICATIONS

1. **Debnath, A.,** Venot, T., Corsi, M-C. & Verma, P. (2025). ***"Neural mechanisms of training in Brain-Computer Interface: A Biophysical modeling approach,"****(under review) (https://doi.org/10.1101/2025.06.21.660834)*
2. **Debnath, A.** (2026). ***"Spectral Graph modeling of abnormal neural synchronizations in Alzheimer's disease."*** (MS Thesis, UCSF)
3. **Debnath, A.***, Barron, N.*, Prabhu, P.,...,Nagarajan, S., & Raj, A., ***"Altered frequency-dependent neural driving signals significantly impair E/I dynamics in Alzheimer's disease."*** *(in prep.)*

SELECT PROJECTS

Decoding Cognitive tasks from functional-MRI using machine learning UCSF | 2026

- Designed ML pipelines to preprocess/process voxel-level task-based fMRI data of two sessions from 10 healthy subjects using fMRIPrep.
- Trained a Support Vector Machine (SVM) classifier on these voxel-level preprocessed BOLD signals to decode and classify different language and motor tasks across 10 healthy subjects. Trained SVM achieved an overall accuracy of 86.1% across tasks and subjects, with a strong performance in decoding language tasks, achieving an F1-score of 0.95, and moderate performance in classifying motor tasks, with an F1-score of 0.77. Notably, 21% of motor task instances were misclassified as language tasks.

Deep learning models for brain MRI denoising: A comparative benchmark study UCSF | 2026

- Designed and implemented deep learning pipelines to train and benchmark denoising autoencoders (CAE, VAE, ResNet, and UNet) using 7000 T1 MRI slices from 128 subjects, under both fixed and variable noise conditions.
- ResNet and UNet consistently achieved strong performance, with SSIM values reaching $\sim$ 0.92–0.96, PSNR $\sim$ 30–34 dB, and EPI $\sim$ 0.85–0.92 across noise levels. In contrast, CAE demonstrated moderate performance (SSIM $\sim$ 0.85–0.91, PSNR $\sim$ 27–31 dB, EPI $\sim$ 0.78–0.88), while VAE showed the lowest scores (SSIM $\sim$ 0.75–0.88, PSNR $\sim$ 25–29 dB, EPI $\sim$ 0.70–0.82) with higher variability.

SKILLS

- **Programming:** Python, MATLAB, C/C++, SQL, LaTeX
- **Packages & Softwares:** NumPy, Pandas, SciPy, Pingouin, Statsmodels, Matplotlib, MNE, Nilearn, fMRIPrep, XCP-D, MRtrix3
- **ML/DL Libraries/Frameworks:** Scikit-Learn, TensorFlow, PyTorch
- **Neuroimaging & Reserch:** Signal/image Processing: EEG, MEG, f-MRI, d-MRI; Neuroscience, BCI, Drug Delivery Systems.
- **Others:** Git/Github, Linux, High Performance Computing (CoreHPC)

SELECTED CONFERENCE POSTERS

1. **Debnath, A.,** Barron, N., Raj, A. (2026). **Spectral Graph modeling of abnormal neural synchronizations in Alzheimer's disease.** *2026 UCSF Annual Radiology Research Conference, Santa Cruz, CA. Poster: March 30, 2026.*
2. **Debnath, A.,** Verma, P., Corsi, M-C. (2025). **A biophysical model to infer neural mechanisms of motor imagery in brain-computer interface training.** *OHBM 2025 Annual Meeting, Brisbane, Australia. Poster: June 27-28, 2025.*
3. **Debnath, A.,** Corsi, M-C., Verma, P. (2025). **A Neural mass modeling approach for uncovering neural mechanisms in motor imagery-based brain-computer interface.** *Wadhwani School of Data Science and Artificial Intelligence (WSAI) Annual Research Showcase 2025, IIT Madras, India. Poster: May 17, 2025.*
4. **Debnath, A.,** Corsi, M-C., Verma, P. (2025). **Biophysical modeling to elucidate the underlying neural mechanisms in EEG-based brain-computer interface.** *Physics of Cells & Tissues (PoCT) 3.0 Symposium, Indian Institute of Science (IISc), Bengaluru, India. Poster: February 21, 2025.*

AWARDS AND HONORS

- MS Thesis Research Grant ($8000 in total) Summer 2026
- IBSE Post-Baccalaureate Fellowship ($6000 in total) 2024–2025
- Paris Brain Institute Travel Award for OHBM 2025 Annual Meeting at Brisbane 2025
- IBSE Travel Award for Physics of Cells & Tissues 3.0 Symposium at IISc 2025
- NEC Merit Scholarship to pursure Bachelor's degree ($1140 in total) 2020–2024

RELEVANT COURSES TAKEN

Graduate Courses:

- **Biomedical Imaging/Bioengineering & Neuroscience:** Principles of MRI – MRI Physics, Systems and Behavioural Neuroscience, Advanced Neurological Imaging, Machine Learning Algorithms for Medical Imaging, Image Processing and Analysis-I & II, Physical Principles of CT, PET and SPECT Imaging, Ultrasound, Imaging Probes for Nuclear and Optical Imaging, Imaging Study Design.

UG Courses:

- **Biotechnology:** Cell Biology and Genetics, Molecular Biology and rDNA Technology, Biochemistry and Enzyme Technology, Microbiology and Bioprocess Technology, Human Genomics, Bioinformatics, Cancer Biology and Cell Signaling, Applications of Molecular Cloning.
- **Mathematics:** Linear Algebra, Calculus, Ordinary Differential Equations, Probability, Partial Differential Equations.
- **Computer Science:** Introduction to Computing, Programming and Data Structures, Compiler Design, Database Management Systems.
- **Others:** Electrical Eng.– Control Systems.